USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 23 LAB - Grating Lobes, Squint, and Quantization
**Lab Sheet**
Fall 2026

**Name:** _____ **Section:** _____ **Date:** _____

**Learning Objective 3.8:** I can identify beam squint and quantization effects in a phased array and describe their impact on system performance.

This sheet is the turn-in document for the L23 Lab. Lesson 23 derived the three departures no taper can fix; here you measure them. Predict each row before you sweep it, and use the Signal Freq the GUI shows for $\lambda$ throughout (about 10.5 GHz, so $\lambda \approx 28.5$ mm). The lab is individual: one sheet per cadet, from your own kit.

1. **Grating lobes.** Source at boresight, Enforce Symmetric Taper off. Lab preset 4 leaves Rx1, Rx4, and Rx7 on, so the active spacing is $d_{\text{eff}} = 3d = 42$ mm. Then set Rx4 and Rx7 to 0 and Rx5 to 100 for $d_{\text{eff}} = 4d = 56$ mm. A full-height lobe appears at every angle where $\sin\theta = m\lambda/d_{\text{eff}}$ has a real solution. Predict the angles before each sweep. The peak drop is read against the frozen uniform trace, and its calculated value follows from the number of elements still on.

| Elements on | $d_{\text{eff}}$ | Lobes calc | Lobes meas | Height vs $0^\circ$ lobe | Drop calc | Drop meas |
|---|---|---|---|---|---|---|
| All eight | 14 mm | | | | | |
| Rx1, Rx4, Rx7 | 42 mm | | | | | |
| Rx1, Rx5 | 56 mm | | | | | |

   In one sentence: every grating lobe stands as tall as the main lobe, and no sidelobe ever does. What is the difference between the two?

   _____

   _____

   _____

   _____

2. **Beam squint.** Move the HB100 to about $45^\circ$ on the protractor arc and press the Uniform preset. Lab preset 5 loads Signal BW = 500 MHz: the GUI computes the steering phases for a frequency 500 MHz below the Signal Freq and measures at the Signal Freq, which is what a signal at the edge of a 500 MHz band sees. Set Signal BW to 10 MHz, sweep, read Est. Angle, and press Freeze. Then set it back to 500 MHz and sweep again. The calculated shift comes from the squint relation in Lesson 23: a beam commanded to $\theta_0$ with its phases set at $f_0$ points at $\sin^{-1}[(f_0/f)\sin\theta_0]$ when observed at $f$.

| Signal BW | Est. Angle meas | Shift calc | Shift meas |
|---|---|---|---|
| 10 MHz | | | |
| 500 MHz | | | |

   The source did not move between the two sweeps. In one sentence, what did?

3. **Phase quantization.** Source back at boresight. Lab preset 6 loads the Blackman taper, so every true sidelobe is below the floor and anything that appears is the phase shifter's doing. Turn **Use Bits** off, so the sweep keeps stepping the commanded angle by $2.8125^\circ$ while the Phase Shift Bits slider coarsens the element phases. Sweep at 7, 4, 3, and 2 bits. The Blackman main lobe is about $24^\circ$ wide and its skirt reaches the floor near $\pm 35^\circ$, so record the highest lobe *beyond* $\pm 35^\circ$ relative to the peak, and beside it the Lesson 23 rule of thumb for the quantization sidelobe level, $-6B$ dB.

| **Bits $B$** | **LSB calc** | **Highest lobe beyond $\pm 35^\circ$, meas** | **$-6B$ dB** |
|---|---|---|---|
| 7 | | | |
| 4 | | | |
| 3 | | | |
| 2 | | | |

The rule of thumb puts the 7-bit lobes near $-42$ dB. In one sentence, why does that row have no number, and what is the deepest level this sweep could report for this taper?

Before you leave: Use Bits on, Phase Shift Bits 7, Signal BW 10 MHz, and the Uniform preset.

**Documentation:** _____